Notes: Boutchkova, Doshi, Durnev, and Molchanov (RFS, 2012)

Precarious Politics and Return Volatility

Contents

1 Big picture

- **Question.** Do domestic and foreign political risks increase stock-return volatility, and do some industries react more strongly to political risk than others?
- **Setting.** The paper studies publicly listed firms in **50 countries** over **1990–2006**. The unit of analysis is mainly a two-digit SIC **industry–country–year** observation.
- **Core idea.** Political events do not affect all industries equally. The paper therefore measures industries' **political sensitivity** along three dimensions:
  - dependence on **international trade**,
  - dependence on **contract enforcement**, and
  - **labor intensity**.
- **Main empirical design.** The paper interacts industry-level political sensitivities with country-level political variables such as elections, autocracy, political risk, left-party government, and labor-law rigidity.
- **Why industry-level analysis matters.** Aggregate country regressions often fail to detect the political-volatility channel because political shocks have heterogeneous effects across sectors. The paper's design asks whether politically exposed sectors become more volatile *relative to other sectors in the same broad environment.*
- **Main results.**
  - More export-oriented industries are more volatile when domestic or foreign political risk is high.
  - Foreign political risk matters especially for **idiosyncratic** volatility of trade-dependent industries.
  - Domestic political risk matters especially for **systematic** volatility.
  - Industries that rely more on complex input relationships, and hence on contract enforcement, are more volatile during domestic elections and high political risk.
  - Labor-intensive industries are more volatile under left governments and in countries with stricter labor regulations.
  - Autocratic regimes tend to dampen volatility for industries exposed to trade and contract enforcement, consistent with a "stability" effect of entrenched political arrangements.
- **Economic takeaway.** Political risk is not just a country-level background variable. It is transmitted into asset-market volatility through concrete industry channels: trade links, contracting needs, and labor regulation.

2 Data and measurement (key objects)

2.1 Observed firm, industry, and country data

- **Stock-return data.** Firm return data come from **Datastream**; accounting and industry information come from **Worldscope**.

- **Sample period.** The main volatility sample covers **1990–2006**.
- **Country coverage.** The paper studies firms in **50 countries**, including both developed and developing economies.
- **Industry definition.** Industries are defined at the **two-digit SIC** level. The primary sample contains **57 SIC industries**.
- **Firm inclusion rule.** A firm must have at least **26 weeks** of return data in a year.
- **Outlier treatment.** Weekly returns below -75% or above $+75\%$ are dropped. Volatility measures are winsorized at the 1st and 99th percentiles.
- **Scale of the dataset.** The main return-volatility sample contains **26,429 industry–country–year observations**, based on **286,433 firm-year observations**.

2.2 Return volatility measures

- The dependent variable is the log of annualized industry return volatility.
- For firm f, weekly return volatility is computed as

$$\sigma_f = \left[\frac{1}{W-1} \sum_{w=1}^{W} \left(r_{f,w} - \frac{1}{W} \sum_{w=1}^{W} r_{f,w} \right)^2 \right]^{1/2}.$$

- Industry volatility is the average of firm volatilities within an industry:

$$\sigma_{ind} = \frac{1}{F_{ind}} \sum_{f=1}^{F_{ind}} \sigma_f.$$

- The paper also decomposes volatility into systematic and idiosyncratic parts using a two-factor model:

$$r_{f,w} = \alpha_f + \beta_{1,f} r_{world,w} + \beta_{2,f} r_{country,w} + \varepsilon_{f,w}.$$

- **Systematic volatility** is the standard deviation of the explained component of returns.
- **Idiosyncratic volatility** is the standard deviation of the residual component $\varepsilon_{f,w}$.
- For election years, the volatility window is centered on the election date: 26 weeks before and 26 weeks after the election.

2.3 Industry political sensitivity measures

- **International trade exposure.** This is the share of industry sales exported to foreign markets. Export data come from UNCTAD/WTO PC-TAS, and sales data come from the United Nations Industrial Commodity Production Statistical database.
- For industry *ind* in country c, exports to trading partner j in year t are scaled by sales:

$$\text{TRADE}^c_{ind,j,t} = \frac{\text{EXP}^c_{ind,j,t}}{\text{SALES}^c_{ind,t}}.$$

- Total export exposure is
$$\text{TRADE}^c_{ind,t} = \sum_j \text{TRADE}^c_{ind,j,t}.$$
- **Contract enforcement sensitivity.** This measures how complex an industry's input structure is. The idea is that industries using many different inputs rely more on contract enforcement.
- The measure is one minus the Herfindahl index of industry input shares:
$$C_{ind} = 1 - \sum_k \phi^2_{ind,k},$$
where $\phi_{ind,k}$ is the share of input from industry k used by industry ind.
- **Labor intensity.** This is the value of labor inputs divided by total input value:
$$LI_{ind,t} = \frac{v^l_{ind,t}}{v^l_{ind,t} + v^k_{ind,t} + v^e_{ind,t} + v^m_{ind,t}},$$
where the denominator includes labor, capital services, energy, and materials.
- Contract enforcement sensitivity and labor intensity are mostly constructed from U.S. input-output or production data and then applied across countries. This is meant to capture technological features of industries rather than country-specific institutional responses.

2.4 Country political variables

- **National elections.** A dummy equal to one in years with presidential elections in presidential systems or parliamentary elections in parliamentary systems.
- **Autocracy.** Based on the POLITY score. The paper transforms the original scale so that higher values mean more autocratic regimes.
- **Political risk.** Based on the International Country Risk Guide index. The paper transforms the index so that higher values mean greater political risk.
- **Party orientation.** A dummy equal to one when the chief executive's party is classified as left-wing.
- **Rigidity of employment.** A World Bank Doing Business index, available for 2004–2006, where higher values indicate more rigid employment regulation.

2.5 Descriptive facts

- Average trade exposure across countries, industries, and years is about **5.44%**.
- Some industries are highly trade-exposed: electronic and electrical equipment has trade exposure of about **16.64%**; industrial and computer equipment has about **9.57%**.
- Some industries have very low trade exposure: building materials is about **2.41%**, and electric, gas, and sanitary services is about **2.28%**.
- Average contract enforcement sensitivity is about **0.841**. Building construction has a high score because it uses many different inputs.
- Average labor intensity is about **0.275**. Petroleum refining is very low at about **0.057**; measuring instruments is high at about **0.501**; hotels are also high at about **0.481**.

2.6 Controls and timing

- The regressions include industry, country, and year fixed effects.
- Country controls include GDP per capita, law-and-order quality, exchange-rate risk, financial development, and ownership concentration.
- Industry controls include size, leverage, diversification, equity dependence, and skill dependence.
- The paper also includes interactions between political variables and relevant controls to reduce concerns that political sensitivity is proxying for economic development, legal quality, or exchange-rate instability.

3 Models and empirical specifications (all equations + interpretation)

3.1 Main panel specification

The main regression is

$$LVOL^{c}_{ind,t} = \alpha_{ind} + \delta_c + \eta_t + \beta\left(\text{SENSITIVITY}^{c}_{ind,t} \times \text{POLITICAL}^{c}_{t}\right) + \gamma\text{POLITICAL}^{c}_{t} + \lambda\text{SENSITIVITY}^{c}_{ind,t} + \text{CONT}$$

- $LVOL^{c}_{ind,t}$ is the log of annualized volatility for industry ind, country c, and year t.
- α_{ind} are industry fixed effects.
- δ_c are country fixed effects.
- η_t are year fixed effects.
- SENSITIVITY is one of the industry exposure measures: trade exposure, contract enforcement sensitivity, or labor intensity.
- POLITICAL is one of the political variables: elections, autocracy, political risk, left-party orientation, or labor-law rigidity.
- The coefficient of interest is β.

Interpretation. The coefficient β asks: when a country experiences a given political condition, do industries that are more exposed to that political channel become differentially more volatile?

3.2 Trade exposure and domestic politics

For domestic politics, the key interaction is

$$\text{TRADE}^{c}_{ind,t} \times \text{POLITICAL}^{c}_{t}.$$

- If $\beta > 0$, trade-dependent industries are more volatile when domestic political uncertainty is higher.
- A positive coefficient can arise because export subsidies, licenses, trade policy, or regulatory support become more uncertain around elections or in risky political environments.

- A negative coefficient with autocracy can arise if autocratic governments provide stable support to favored export sectors.

Interpretation. Export exposure is not only an international-demand channel. It can also make firms sensitive to domestic political decisions that affect their ability to sell abroad.

3.3 Trade exposure and global politics

The paper also constructs a trading-partner-weighted global political exposure:

$$\text{GLOBALPOL}^{c}_{ind,t} = \sum_{j} \left(\text{TRADE}^{c}_{ind,j,t} \times \text{POLITICAL}_{j,t} \right).$$

- This variable is large when an industry exports heavily to countries that are experiencing elections, high political risk, or autocracy.
- It separates foreign political risk from home-country political risk.
- The idea is that firms relying on exports to politically risky destinations face more uncertain foreign demand, payment, regulation, or contract conditions.

Interpretation. Trade can diversify away domestic demand risk, but it also imports political risk from trading partners.

3.4 Contract enforcement channel

The contract-enforcement specification uses

$$C_{ind} \times \text{POLITICAL}^{c}_{t},$$

where

$$C_{ind} = 1 - \sum_{k} \phi^{2}_{ind,k}.$$

- A high value of C_{ind} means that production uses inputs from many different supplying industries.
- Such industries are more vulnerable to weak institutions because more supplier relationships must be governed by contracts.
- Elections and political risk may disrupt contract enforcement or change the perceived stability of institutional arrangements.

Interpretation. Political risk affects firms not only through demand or policy but also through the reliability of contracting relationships inside the production process.

3.5 Labor intensity channel

The labor-channel regressions use interactions such as

$$LI_{ind,t} \times \text{LeftParty}^{c}_{t},$$

$$LI_{ind,t} \times \text{Election}^{c}_{t},$$

and

$$LI_{ind,t} \times \text{EmploymentRigidity}_t^c.$$

- Left parties are expected to support more labor-friendly and redistributive policies.
- Stricter labor regulation can make it harder for firms to adjust employment after negative shocks.
- Therefore labor-intensive industries should have more volatile cash flows and stock returns when labor policy is more restrictive or uncertain.

Interpretation. The political risk here operates through factor-market rigidity: if labor adjustment is costly, negative shocks have a larger effect on profits.

3.6 Volatility decomposition

The paper estimates all main specifications using three dependent variables:

$$LVOL_{ind,c,t}^{Total}, \quad LVOL_{ind,c,t}^{Idio}, \quad LVOL_{ind,c,t}^{Sys}.$$

- **Total volatility** captures the overall instability of industry returns.
- **Idiosyncratic volatility** captures volatility not explained by world and country market factors.
- **Systematic volatility** captures the component explained by global and domestic market indexes.

Interpretation. This decomposition is central because the paper finds that global political risk is mainly connected to idiosyncratic volatility, while domestic political risk is more connected to systematic volatility.

3.7 Endogeneity and robustness specifications

The paper addresses several endogeneity concerns:

- **Autocracy and political risk.** The authors use instrumental variables based on settlers' mortality and ethnolinguistic fractionalization.
- **Elections.** They separate fixed-timing elections from flexible-timing and early elections to check whether results are driven by endogenous election timing.
- **Left-party orientation.** They decompose left-party orientation into an explained part and an unexplained part using lagged macro variables.
- **Political crises.** They interact industry sensitivities with a political-crisis dummy as another test of political shocks.
- **Alternative volatility measures.** They reweight firm volatility by assets, debt, and diversification, and also examine pure-play firms.
- **Fundamental volatility.** They control for ROA volatility to check whether results are purely driven by volatility of fundamentals.

- **Alternative labor intensity.** They use country-specific labor intensity from Worldscope, measured as employees divided by sales.

Interpretation. The robustness tests are designed to show that the results are not just artifacts of country-level omitted variables, election timing, volatility construction, or U.S.-based industry measures.

4 Identification and instruments (what makes the estimates credible)

4.1 Why aggregate country regressions are not enough

- A simple regression of country volatility on elections or political risk is hard to interpret because political risk is correlated with macroeconomic conditions, financial development, institutions, and crises.
- The paper shows that some aggregate comparisons are weak: average volatility is not significantly different during election years versus nonelection years, nor under left governments versus other governments.
- Aggregate country-level tests can miss political effects if only some industries are exposed to the relevant political channel.

4.2 Rajan–Zingales-style industry sensitivity design

- The identifying variation comes from whether politically sensitive industries become more volatile when the relevant country political variable is high.
- Industry fixed effects absorb time-invariant differences across industries, such as naturally high volatility in technology or mining.
- Country fixed effects absorb time-invariant differences across countries, such as persistent differences in financial development or legal origins.
- Year fixed effects absorb global shocks common to all countries and industries.
- The interaction design therefore focuses on **differential within-country industry responses** to political risk.

Core identifying assumption. Conditional on controls and fixed effects, industries with higher measured political sensitivity would not have become differentially more volatile in politically risky country-years for reasons unrelated to the proposed political channel.

4.3 Why the three channels are informative

- **Trade exposure** identifies exposure to domestic and foreign political uncertainty because export-dependent industries rely on cross-border regulations, foreign demand, and trading-partner stability.

- **Contract enforcement sensitivity** identifies exposure to institutional quality because input-complex industries rely on many supplier relationships.
- **Labor intensity** identifies exposure to labor politics because labor-intensive industries are more affected by firing costs, hiring restrictions, and redistributive labor policy.

4.4 Domestic versus global political risk

- A distinctive feature of the paper is that it separates home-country political risk from trading-partner political risk.
- This is possible because the authors observe industry-level exports to specific destinations.
- Foreign political variables are weighted by export shares, so the same home-country industry can have different global political exposure depending on where it sells its output.

Interpretation. This lets the paper show that trade is not simply a diversification device. It can transmit foreign political shocks into domestic equity volatility.

4.5 Instrumental variables and endogeneity checks

- For autocracy and political risk, the paper uses **settlers' mortality** and **ethnolinguistic fractionalization** as instruments.
- For election timing, the paper checks whether results differ between fixed-timing elections, flexible-timing elections, and early elections.
- For left-party orientation, the paper first predicts left-party government using lagged macro variables and then studies whether the unexplained component still matters.
- For political crises, the paper uses a crisis dummy based on 46 political disruptions across 18 countries.

Interpretation. These checks do not turn the design into a perfect experiment, but they address the most obvious concerns that economic volatility itself may cause political variables.

4.6 Limitations

- The paper does not include country-year fixed effects because the political variables vary at the country-year level and would be absorbed.
- Some omitted country-year shocks could still matter if they affect high-sensitivity industries differently.
- Industry sensitivities measured from U.S. data may not perfectly describe production technologies in all countries.
- The paper studies volatility, not expected returns, so it does not directly test whether political risk is priced.

4.7 Relation to the literature

- The paper contributes to the literature on politics and financial markets by showing that routine political processes, not just coups, wars, and crises, affect return volatility.
- It contributes to international finance by showing that trade linkages transmit foreign political risk.
- It contributes to the industry-level finance literature by identifying business characteristics that shape systematic and idiosyncratic volatility.

5 Tables and figures

5.1 Table 1: variables, definitions, and sources

Read:

- Table 1 is the measurement map of the paper.
- It defines the three volatility outcomes: total, idiosyncratic, and systematic volatility.
- It also defines the three industry political-sensitivity measures: international trade exposure, contract enforcement sensitivity, and labor intensity.
- The political variables are carefully transformed so that higher values mean more autocracy or more political risk.
- The table also lists the main controls, including exchange-rate risk, law efficiency, GDP per capita, industry size, leverage, diversification, equity dependence, financial development, skill dependence, and ownership concentration.

5.2 Table 2: descriptive statistics by industry and country

Table 2
Descriptive statistics

Panel A: Descriptive statistics by industry

Industry name	SIC code	Total volatility	Idiosyncratic volatility	Systematic volatility	International trade exposure (%)	Sensitivity to contract enforcement	Labor intensity	Number of country-years	Number of firm-years
Agricultural crops	100	0.444	0.408	0.148	5.03	–	0.245	319	1, 130
Agriculture livestock	200	0.446	0.415	0.143	2.76	0.899	–	510	1, 308
Forestry	800	0.389	0.355	0.136	–	0.495	–	292	904
Fishing and hunting	900	0.605	0.569	0.180	3.17	–	–	136	533
Metal mining	1000	0.556	0.505	0.196	3.40	0.905	0.183	415	8, 481
Coal mining	1200	0.525	0.474	0.188	2.58	–	0.291	228	1, 011
Oil and gas extraction	1300	0.511	0.456	0.194	7.47	0.799	0.132	415	6, 317
Quarrying of minerals	1400	0.498	0.452	0.179	4.66	0.913	0.283	311	1, 720
Building construction	1500	0.448	0.400	0.177	–	0.933	0.369	1, 299	9, 743
Food products	2000	0.384	0.340	0.154	6.75	0.860	0.174	753	11, 281
Tobacco products	2100	0.359	0.318	0.147	3.26	–	0.154	282	1, 015
Textile mill products	2200	0.472	0.427	0.171	5.47	0.823	0.250	562	5, 041
Apparel	2300	0.446	0.403	0.165	7.29	–	0.303	587	2, 809
Lumber and wood products	2400	0.446	0.402	0.169	4.01	0.813	0.257	408	1, 503
Furniture and fixtures	2500	0.461	0.422	0.160	3.04	0.923	0.372	433	1, 928
Paper and allied products	2600	0.419	0.368	0.174	4.41	0.861	0.255	621	4, 126
Printing and publishing	2700	0.438	0.396	0.159	–	0.931	0.423	552	3, 892
Chemicals and allied products	2800	0.456	0.406	0.179	7.81	0.832	0.195	701	19, 467
Petroleum refining	2900	0.402	0.321	0.211	4.36	0.512	0.057	440	1, 996
Rubber and plastics products	3000	0.462	0.409	0.187	6.28	0.826	0.333	561	4, 083

Table 2
Continued

Panel A: Descriptive statistics by industry

Industry name	SIC code	Total volatility	Idiosyncratic volatility	Systematic volatility	International trade exposure (%)	Sensitivity to contract enforcement	Labor intensity	Number of country-years	Number of firm-years
Leather and leather products	3100	0.452	0.400	0.185	3.8	0.830	0.245	262	1, 201
Stone, clay, and glass	3200	0.436	0.376	0.189	8.14	–	0.344	669	5, 410
Primary metal industries	3300	0.458	0.401	0.192	7.77	0.791	0.195	663	7, 853
Fabricated metal products	3400	0.458	0.413	0.168	3.98	0.847	0.303	517	4, 928
Industrial and computer equipment	3500	0.490	0.443	0.181	9.57	0.910	0.330	630	13, 960
Electronic and electrical equipment	3600	0.506	0.452	0.194	16.64	0.909	0.274	607	18, 629
Transportation equipment	3700	0.441	0.391	0.176	8.89	0.819	0.139	602	7, 279
Measuring instruments	3800	0.468	0.425	0.173	4.91	0.843	0.501	428	8, 799
Miscellaneous industries	3900	0.490	0.449	0.171	4.17	0.939	0.272	437	2, 811
Railroad transportation	4000	0.315	0.281	0.124	3.32	0.913	0.354	259	917
Highway passenger Transportation	4100	0.392	0.360	0.135	–	0.935	–	292	1, 282
Motor freight transportation	4200	0.413	0.373	0.155	–	0.896	–	393	2, 361
Water transportation	4400	0.417	0.373	0.163	–	0.851	–	574	2, 990
Transportation by air	4500	0.452	0.395	0.191	–	0.852	–	510	1, 909
Pipelines, except natural gas	4600	0.365	0.338	0.121	–	0.884	–	134	724
Transportation services	4700	0.463	0.413	0.178	–	0.930	–	499	2, 477
Communications	4800	0.471	0.386	0.233	–	0.796	0.223	637	5, 818
Electric, gas, and sanitary services	4900	0.393	0.330	0.184	2.28%	0.807	0.191	642	7, 643
Wholesale trade-durable goods	5000	0.469	0.421	0.179		0.913	0.453	626	10, 680
Building materials	5200	0.424	0.388	0.146	2.41%		–	235	1, 013
Eating and drinking places	5800	0.443	0.401	0.164	–	0.847	–	367	2, 908

Read:

- Panel A shows large cross-industry variation in volatility and political sensitivities.
- Trade exposure is especially high in electronic and electrical equipment, and low in utilities and building-material sectors.
- Contract enforcement sensitivity is high for industries with complex input structures, such as construction.

- Labor intensity is highest in industries such as measuring instruments and hotels, and lowest in petroleum refining.
- Panel B shows cross-country variation in GDP per capita, volatility, autocracy, and political risk.
- The descriptive variation is important because the empirical design needs industries and countries to differ meaningfully in the relevant exposure measures.

Table 2
Continued

Panel A: Descriptive statistics by industry

Industry name	SIC code	Total volatility	Idiosyncratic volatility	Systematic volatility	International trade exposure (%)	Sensitivity to contract enforcement	Labor intensity	Number of country-years	Number of firm-years
Miscellaneous retail	5900	0.477	0.432	0.165	–	0.922	–	477	3, 530
Depository institutions	6000	0.355	0.285	0.189	–	0.897	0.223	768	15, 984
Security and commodity brokers	6200	0.444	0.384	0.190	–	0.819	–	596	4, 676
Insurance carriers	6300	0.364	0.304	0.176	–	0.372	–	559	5, 005
Real estate	6500	0.427	0.377	0.173	–	0.920	–	651	10, 511
Investment offices	6700	0.436	0.385	0.172	–	0.382	–	644	12, 089
Hotels	7000	0.437	0.389	0.174	–	0.946	0.481	510	2, 636
Business services	7300	0.537	0.489	0.194	–	0.928	–	612	21, 625
Motion pictures	7800	0.545	0.502	0.188	–	0.743	–	263	1, 530
Amusement services	7900	0.477	0.436	0.162	–	0.921	–	457	2, 998
Health services	8000	0.504	0.463	0.171	–	0.936	–	390	2, 682
Legal services	8100	0.649	0.610	0.185	–	0.892	–	86	417
Educational services	8200	0.556	0.517	0.178	–	0.884	–	266	1, 328
Social services	8300	0.496	0.472	0.131	–	0.939	–	142	662
Museums and art galleries	8400	0.385	0.369	0.093	–	0.819	–	120	469
Membership organizations	8600	0.516	0.471	0.195	–	0.883	–	80	411
Average		0.456	0.409	0.171	5.44%	0.841	0.275	464	5, 025
Minimum		0.315	0.281	0.093	2.28%	0.372	0.057		
Maximum		0.649	0.610	0.233	16.64%	0.946	0.501		
Total								26, 429	286, 433
Number of industries		57	57	57	29	50	32		

This table contains summary statistics by industry (average values across countries and years from 1990 to 2006). SIC code is two-digit Standard Industry Classification code. All of the variables are defined in Table 1. *Number of country-years* is the aggregate number of country observations across industries and sample years. *Number of firm-years* is the aggregate number of firm observations across industries and sample years used to calculate the volatility measures. The last row contains the number of industries for each variable.

Table 2
Continued

Panel B: Descriptive statistics by country

Country	GDP per capita (in 2000 US$)	Total volatility	Idiosyncratic volatility	Systematic volatility	Availability of return series	Autocracy	Political risk	Number of industry-years	Number of firm-years
Argentina	7, 334	0.497	0.398	0.268	1990–2006	2.333	30.147	318	774
Australia	19, 854	0.468	0.432	0.159	1990–2006	0.000	15.647	982	12, 827
Austria	22, 684	0.309	0.269	0.136	1990–2006	0.000	13.559	386	961
Belgium	21, 384	0.323	0.292	0.118	1990–2006	0.000	18.412	493	1, 497
Brazil	3, 641	0.621	0.524	0.291	1990–2006	2.000	34.206	580	2, 351
Canada	21, 989	0.516	0.490	0.139	1990–2006	0.000	15.353	1, 021	13, 788
Chile	4, 572	0.374	0.321	0.178	1990–2006	1.333	25.353	537	1, 961
China	880	0.442	0.424	0.109	1990–2006	17.000	33.706	655	12, 600
Colombia	2, 427	0.453	0.322	0.290	1992–2006	3.000	45.700	137	300
Czech Rep.	5, 507	0.387	0.300	0.210	1994–2006	0.167	20.846	100	184
Denmark	28, 130	0.344	0.307	0.137	1990–2006	0.000	13.088	482	2, 180
Egypt	1, 332	0.469	0.393	0.235	1995–2006	15.500	38.206	93	212
Finland	21, 905	0.374	0.337	0.139	1990–2006	0.000	10.853	500	1, 423
France	21, 358	0.408	0.382	0.122	1990–2006	1.000	21.088	981	8, 167
Germany	22, 012	0.363	0.337	0.115	1990–2006	0.000	16.265	909	9, 317
Greece	11, 329	0.518	0.482	0.158	1990–2006	0.000	24.594	705	3, 321
Hungary	4, 541	0.495	0.429	0.197	1991–2006	0.000	21.706	171	310
India	433	0.559	0.497	0.231	1990–2006	1.000	42.588	594	5, 959
Indonesia	811	0.602	0.526	0.251	1990–2006	7.667	45.324	650	3, 177
Ireland	21, 891	0.401	0.373	0.124	1990–2006	0.000	14.882	336	801
Israel	17, 998	0.484	0.436	0.188	1990–2006	0.333	38.235	413	1, 788
Italy	18, 309	0.358	0.310	0.160	1990–2006	0.000	23.588	623	2, 803
Japan	36, 289	0.425	0.369	0.189	1990–2006	0.000	17.441	1, 045	45, 751
Luxembourg	42, 342	0.344	0.319	0.113	1991–2006	0.000	7.588	236	371
Malaysia	3, 765	0.480	0.397	0.249	1990–2006	7.000	27.412	907	10, 009
Mexico	5, 547	0.398	0.348	0.161	1990–2006	3.167	29.429	384	1, 057
Morocco	1, 338	0.287	0.238	0.140	1993–2006	16.250	32.353	93	255
Netherlands	22, 286	0.344	0.310	0.129	1990–2006	0.000	12.059	594	2, 209

(*continued*)

Table 2
Continued

Panel B: Descriptive statistics by country

Country	GDP per capita (in 2000 US$)	Total volatility	Idiosyncratic volatility	Systematic volatility	Availability of return series	Autocracy	Political risk	Number of industry-years	Number of firm-years
New Zealand	12, 805	0.417	0.385	0.138	1990–2006	0.000	14.559	584	1, 150
Norway	34, 932	0.423	0.382	0.158	1990–2006	0.000	13.618	427	1, 471
Pakistan	530	0.470	0.422	0.176	1991–2006	11.083	52.382	289	1, 208
Peru	1, 996	0.358	0.337	0.096	1991–2006	5.000	41.188	183	484
Philippines	959	0.541	0.505	0.162	1990–2006	2.000	36.719	472	2, 227
Poland	4, 034	0.452	0.383	0.215	1991–2006	0.583	24.265	281	803
Portugal	10, 086	0.356	0.319	0.135	1990–2006	0.000	14.385	381	716
Russia	1, 987	0.551	0.403	0.328	1994–2006	5.250	40.833	131	360
Singapore	20, 860	0.479	0.430	0.183	1990–2006	12.000	16.324	785	4, 973
South Africa	3, 095	0.486	0.438	0.187	1990–2006	1.000	32.235	647	3, 490
South Korea	10, 565	0.566	0.494	0.248	1990–2006	2.500	24.912	831	10, 319
Spain	13, 452	0.337	0.297	0.141	1990–2006	0.000	23.000	588	1, 686
Sri Lanka	794	0.469	0.372	0.258	1990–2006	4.750	48.176	140	336
Sweden	26, 118	0.424	0.372	0.184	1990–2006	0.000	13.941	622	2, 951
Switzerland	33, 761	0.330	0.301	0.119	1990–2006	0.000	11.765	595	3, 082
Taiwan	12, 931	0.458	0.369	0.244	1990–2006	1.000	22.324	647	9, 752
Thailand	1, 961	0.515	0.452	0.215	1990–2006	2.167	33.088	778	5, 031
Turkey	3, 881	0.698	0.523	0.434	1990–2006	2.833	41.412	602	2, 748
United Kingdom	23, 294	0.368	0.344	0.111	1990–2006	0.000	16.353	1062	19, 998
United States	32, 454	0.506	0.499	0.144	1990–2006	0.000	17.706	1147	66, 845
Venezuela	4, 939	0.563	0.418	0.341	1990–2006	3.250	40.088	145	239
Zimbabwe	594	0.475	0.470	0.062	1990–2006	14.500	47.353	167	211
Average	12, 958	0.446	0.390	0.184		2.913	26.325	529	5, 729
Minimum	433	0.287	0.238	0.062		0	7.588		
Maximum	42, 342	0.698	0.562	0.434		17	52.382		
Total								26, 429	286, 433

This table contains summary statistics by country (average values across industries and years from 1990 to 2006). All of the variables are defined in Table 1. *Number of industry-years* is the aggregate number of industry observations across countries and sample years. *Number of firm-years* is the aggregate number of firm observations across countries and sample years used to calculate the volatility measures.

5.3 Table 3: mean comparison tests

Read:

- Election years and nonelection years have very similar average total volatility: **0.451** versus **0.448**; the difference is not statistically significant.
- Left-party years and other-party years are also similar: **0.441** versus **0.447**; again not statistically significant.
- High-autocracy observations have lower volatility than low-autocracy observations: **0.422** versus **0.520**.
- High-political-risk observations have higher volatility than low-risk observations: **0.523** versus **0.404**.
- The table motivates the paper's industry-sensitivity approach: aggregate political indicators alone do not reveal the full transmission mechanism.

5.4 Table 4: trade exposure and political variables

Read:

- Table 4 is the main evidence for the trade channel.
- Global elections interacted with trade exposure raise total volatility and idiosyncratic volatility. The coefficients are about **0.4108** and **0.3986**, both highly significant.
- National elections interacted with trade exposure raise total and systematic volatility. This suggests that domestic election risk enters the market-wide component more strongly.
- Global political risk interacted with trade exposure raises total and idiosyncratic volatility. The main coefficients are about **0.0816** for total volatility and **0.0521** for idiosyncratic volatility.
- National political risk interacted with trade exposure raises total and systematic volatility. The main coefficients are about **0.0650** for total volatility and **0.0513** for systematic volatility.

Table 3
Mean comparison tests of volatility for subsamples based on elections, party orientation, autocracy, and political risk

Panel A			Panel B		
Elections	N	Total volatility	Party orientation	N	Total volatility
ELECTIONS	5,642	0.451	LEFT PARTY	8,370	0.441
NO ELECTIONS	19,902	0.448	OTHER PARTIES	9,362	0.447
DIFFERENCE		0.003	DIFFERENCE		−0.006
t-statistic		0.440	t-statistic		−0.523
p-value		(.408)	p-value		(.302)

Panel C			Panel D		
Autocracy	N	Total volatility	Political risk	N	Total volatility
HIGH AUTOCRACY	6,607	0.422	HIGH RISK	6,607	0.523
LOW AUTOCRACY	6,607	0.520	LOW RISK	6,607	0.404
DIFFERENCE		**−0.098*****	DIFFERENCE		**0.119*****
t-statistic		**−3.408**	t-statistic		**3.903**
p-value		(.00)	p-value		(.00)

This table reports the mean comparison tests (with unequal variances) of total return volatility between the sets of observations split according to elections, party orientation, autocracy, and political risk. For these tests, we use the entire sample of industry-country-year observations. The groups are: elections vs. no elections (Panel A); left party versus other parties (Panel B); high autocracy versus. low autocracy (Panel C); and high political risk vs. low political risk (Panel D). The numbers in parentheses are p-values. The coefficients significant at the 10% level (based on a two-tailed test) or higher are in boldface. All of the variables are defined in Table 1. N is the number of observations in each group. The high autocracy (low autocracy) group contains industry-country-year observations that belong to the top 75th percentile, autocracy >2.9 (bottom 25th percentile, autocracy = 0), of the autocracy index. The high political risk (low political risk) group contains industry-country-year observations that belong to the top 75th percentile, political risk >36.7 (bottom 25th percentile, political risk <15.9), of the political risk index.

Table 4
Volatility of industries exposed to international trade conditional on political variables

	Elections			Autocracy			Political risk		
	Total volatility	Iidiosyncratic volatility	Systematic volatility	Total volatility	Idiosyncratic volatility	Systematic volatility	Total volatility	Idiosyncratic volatility	Systematic volatility
Specification	1	2	3	4	5	6	7	8	9
Interaction of International Trade Exposure with Global Political variable	**0.4108*****	**0.3986*****	(0.1101	**−0.0386*****	**−0.0262****	**−0.0430****	**0.0816*****	**0.0521*****	(0.0113
	(0.000)	**(0.000)**	(0.412)	**(0.010)**	**(0.011)**	**(0.038)**	**(0.000)**	**(0.000)**	(0.480)
Interaction of International Trade Exposure with National Political variable	**0.1342*****	0.0480	**0.1162*****	**−0.0617****	(0.0486	**−0.0970*****	**0.0650****	(0.0120	**0.0513*****
	(0.000)	(0.219)	**(0.000)**	**(0.030)**	(0.221)		**(0.011)**	(0.168)	**(0.010)**
Global Political Variable	0.0121	0.0213	0.0314	**−0.1148***	**−0.2119****	−0.0729	**0.0030*****	**0.0018*****	**0.0014*****
	(0.312)	(0.503)	(0.208)	**(0.070)**	**(0.021)**	(0.333)	**(0.000)**	**(0.000)**	**(0.000)**
National Political Variable	**0.1509*****	0.1300	**0.1490***	**−0.0299*****	−0.0068	**−0.0189****	**0.0026***	**0.0020****	**0.0038*****
	(0.000)	(0.142)	**(0.072)**	**(0.010)**	(0.414)	**(0.041)**	**(0.078)**	**(0.031)**	**(0.010)**
Level of International Trade Exposure	**−0.4090*****	**−0.3890*****	**−0.4162*****	**−0.3460*****	**−0.2349*****	**−0.1128*****	**−0.1890*****	**−0.1000***	−0.0804
	(0.000)	**(0.000)**	**(0.000)**	**(0.000)**	**(0.000)**	**(0.000)**	**(0.002)**	**(0.100)**	(0.180)
Interaction of International Trade Exposure with GDP per Capita	−0.0360	−0.0308	−0.0222	−0.0508	−0.0180	−0.0512	−0.0150	0.0032	−0.0108
	(0.281)	(0.312)	(0.390)	(0.222)	(0.309)	(0.581)	(0.314)	(0.600)	(0.295)

(*continued*)

Table 4
Continued

	Elections			Autocracy			Political risk		
	Total volatility	Iidiosyncratic volatility	Systematic volatility	Total volatility	Idiosyncratic volatility	Systematic volatility	Total volatility	Idiosyncratic volatility	Systematic volatility
Specification	1	2	3	4	5	6	7	8	9
Interaction of International Trade Exposure with Efficiency of Law	**−0.2080***	−0.1902	**−0.1421****	**−0.1903*****	**−0.0818***	**−0.1211****	−0.0108	−0.0714	−0.0621
	(0.056)	(0.214)	**(0.050)**	**(0.000)**	**(0.100)**	**(0.050)**	(0.318)	(0.421)	(0.500)
Interaction of International Trade Exposure with Exchange Rate Risk	−0.0514	−0.0621	(0.0208	−0.0706	**−0.0822***	−0.0198	−0.0360	−0.0200	(0.0395
	(0.333)	(0.150)	(0.226)	(0.109)	**(0.100)**	(0.316)	(0.121)	(0.200)	(0.237)
Country GDP per Capita	**−0.4900*****	**−0.2779*****	**−0.3503*****	**−0.4029*****	**−0.3111*****	**−0.2908*****	**−0.3909*****	**−0.2409*****	**−0.2421*****
	(0.000)	**(0.000)**	**(0.000)**	**(0.000)**	**(0.000)**	**(0.000)**	**(0.000)**	**(0.000)**	**(0.000)**
Country Efficiency of Law	**−0.0221*****	**−0.0218***	−0.0130	**−0.0261*****	**−0.0212*****	−0.0139	**−0.0290*****	**−0.0264*****	**−0.0161****
	(0.010)	**(0.052)**	(0.208)	**(0.000)**	**(0.000)**	(0.313)	**(0.000)**	**(0.000)**	**(0.020)**
Country Exchange Rate Risk	**0.0440*****	**0.0333*****	**0.0189*****	**0.0100*****	(0.0086	**0.0293*****	(0.0120	(0.0121	(0.0182
	(0.000)	**(0.000)**	**(0.010)**	**(0.010)**	(0.112)	**(0.000)**	(0.190)	(0.412)	(0.189)
Industry Size	**−0.0129*****	**−0.0093*****	−0.0020	**−0.0176*****	**−0.0172*****	−0.0018	**−0.0230*****	**−0.0289*****	−0.0020
	(0.000)	**(0.000)**	(0.312)	**(0.000)**	**(0.000)**	(0.331)	**(0.000)**	**(0.000)**	(0.412)
Industry Leverage	**0.0738****	(0.0041	**0.0324*****	**0.0830*****	(0.0036	**0.0377****	**0.0690****	(0.00390	**0.0338*****
	(0.040)	(0.312)	**(0.000)**	**(0.000)**	(0.281)	**(0.020)**	**(0.041)**	(0.200)	**(0.000)**
Industry Diversification	**−0.0438*****	**−0.0486*****	−0.0029	**−0.0338*****	**−0.0412*****	0.0042	**−0.0414*****	**−0.0516*****	−0.0011
	(0.000)	**(0.000)**	(0.305)	**(0.000)**	**(0.000)**	(0.200)	**(0.000)**	**(0.000)**	(0.308)

(*continued*)

Table 4
Continued

	Total volatility	Elections Idiosyncratic volatility	Systematic volatility	Total volatility	Autocracy Idiosyncratic volatility	Systematic volatility	Total volatility	Political risk Idiosyncratic volatility	Systematic volatility
Specification	1	2	3	4	5	6	7	8	9
Interaction of Equity Dependence with Financial Development	−0.1960	**−0.2200***	−0.1303	−0.2999	−0.1170	0.0721	−0.1821	**−0.2090***	−0.0614
	(0.140)	**(0.100)**	(0.286)	(0.117)	(0.181)	(0.340)	(0.120)	**(0.100)**	(0.150)
Interaction of Skill Dependence with Ownership Concentration	−0.2030	−0.1030	0.0239	−0.1712	−0.1768	−0.0130	−0.1319	−0.0920	−0.0110
	(0.102)	(0.110)	(0.342)	(0.162)	(0.300)	(0.281)	(0.152)	(0.162)	(0.387)
Equity Dependence	**0.0413***	0.0204	0.0259	**0.0432***	0.0211	0.0186	**0.0309***	**0.0256***	0.0154
	(0.100)	(0.196)	(0.182)	**(0.100)**	(0.101)	(0.289)	**(0.100)**	**(0.060)**	(0.324)
Financial Development	0.0549	0.0262	0.0311	0.0595	0.0288	0.0279	0.0470	0.0172	0.0331
	(0.200)	(0.289)	(0.381)	(0.211)	(0.281)	(0.112)	(0.388)	(0.182)	(0.229)
Skill Dependence	−0.0023	−0.0062	−0.0013	0.0019	−0.0010	0.0011	0.0034	0.0012	−0.0038
	(0.288)	(0.309)	(0.381)	(0.321)	(0.314)	(0.108)	(0.321)	(0.514)	(0.440)
Ownership Concentration	0.0200	0.0123	0.0034	0.0086	0.0017	0.0078	0.0018	0.0036	0.000
	(0.509)	(0.429)	(0.414)	(0.316)	(0.412)	(0.458)	(0.418)	(0.404)	(0.203)
Regression R^2 –adj.	0.4378	0.4294	0.4039	0.4220	0.4118	0.3914	0.4283	0.4262	0.4390
Number of observations	10,633	10,633	10,633	10,633	10,633	10,633	10,633	10,633	10,633

This table reports the results of OLS regressions of the logs of total return volatility, idiosyncratic volatility, and systematic volatility on the interaction terms of industry international trade exposure with political variables: global and national elections index (specifications 1–3), global and national autocracy index (specifications 4–6), and global and national political risk (specifications 7– 9). All of the variables are defined in Table 1. Every regression includes industry, country, and year fixed effects. The numbers in parentheses are p-values. The coefficients significant at the 10% level (based on a two-tailed test) or higher are in boldface. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered by countries and years to adjust for heteroscedasticity, cross-sectional, and time-series correlations.

- Autocracy interactions are negative for trade exposure. The paper interprets this as a stability effect: in some autocratic environments, export relationships may be more predictable for politically supported industries.
- The table's key message is the domestic/global split: foreign politics mainly shows up in idiosyncratic volatility, while domestic politics mainly shows up in systematic volatility.

5.5 Economic magnitudes from the trade regressions

Read:

- The paper compares transportation equipment with leather in Canada and Brazil.
- Using the national political risk coefficient, the relative volatility increase for transportation equipment versus leather is about **19.8%** in Canada and **34.9%** in Brazil.
- Using global political risk, the volatility of Canadian transportation equipment is about **18.7%** larger than leather because transportation equipment exports to riskier destinations.
- A one-standard-deviation increase in the share of trading partners holding elections raises the volatility of Brazil's transportation industry by about **9.8%** relative to leather.
- These magnitudes suggest that routine political events can generate volatility effects comparable in size to more dramatic events documented in earlier literature.

5.6 Table 5: contract enforcement sensitivity

Read:

- Industries more dependent on contract enforcement are more volatile during national elections.
- The election interaction coefficient is **0.0562** for total volatility, **0.0300** for idiosyncratic volatility, and **0.0914** for systematic volatility.
- Contract-sensitive industries are also more volatile when national political risk is high. The political-risk interaction coefficients are positive and significant for all three volatility measures.
- The autocracy interaction is negative and marginally significant for total and systematic volatility.
- The economic magnitude is meaningful. The paper's Canada/Brazil comparison implies that the relative volatility of a high-contract-sensitivity industry versus a lower-sensitivity industry is about **12.0%** larger in Canada but **26.8%** larger in Brazil.
- The table supports the institutional channel: political instability matters more when production depends on many contractual relationships.

Table 5
Volatility of industries sensitive to contract enforcement conditional on political variables

Specification	Total volatility 1	Idiosyncratic volatility 2	Systematic volatility 3
Interaction of Sensitivity to Contract Enforcement with National Elections	**0.0562***	**0.0300***	**0.0914***
	(0.001)	**(0.082)**	**(0.000)**
Interaction of Sensitivity to Contract Enforcement with National Autocracy Index	**−0.0292***	−0.0214	**−0.0321***
	(0.100)	(0.112)	**(0.080)**
Interaction of Sensitivity to Contract Enforcement with National Political Risk Index	**0.0158***	**0.0146***	**0.0200***
	(0.000)	**(0.000)**	**(0.000)**
Sensitivity to Contract Enforcement	**−0.4093***	0.1332	**−0.3003***
	(0.009)	(0.221)	**(0.020)**
National Elections	**0.1403***	**0.1009***	**0.1108***
	(0.076)	**(0.100)**	**(0.009)**
National Autocracy Index	**−0.0216***	**−0.0211***	**−0.0268***
	(0.000)	**(0.000)**	**(0.000)**
National Political Risk Index	**0.0012***	**0.0013***	**0.0015***
	(0.000)	**(0.000)**	**(0.000)**
Interaction of Contract Enforcement Sensitivity with GDP per Capita	**−0.0917***	**−0.0821***	**−0.0432***
	(0.000)	**(0.000)**	**(0.005)**
Interaction of Contract Enforcement Sensitivity with Efficiency of Law	**0.1030***	**0.0940***	**0.0551***
	(0.000)	**(0.000)**	**(0.000)**
Interaction of Contract Enforcement Sensitivity with Exchange Rate Risk	−0.0262	−0.0051	−0.0230
	(0.200)	(0.430)	(0.186)
GDP per Capita	**−0.177***	**−0.1580***	**−0.1621***
	(0.000)	**(0.000)**	**(0.000)**
Efficiency of Law	**−0.0711***	**−0.0701***	**−0.0249***
	(0.000)	**(0.000)**	**(0.000)**
Exchange Rate Risk	**0.0200***	**0.0202***	**0.0213***
	(0.000)	**(0.012)**	**(0.011)**
Industry Size	**−0.0214***	**−0.0221***	0.0070
	(0.000)	**(0.005)**	(0.303)
Industry Leverage	**0.0913***	0.00719	**0.0666***
	(0.012)	(0.211)	**(0.086)**
Industry Diversification	**−0.0443***	**−0.0489***	0.0077
	(0.000)	**(0.000)**	(0.170)
Interaction of Equity Dependence with Financial Development	−0.0926	−0.0182	−0.0071
	(0.282)	(0.333)	(0.232)
Interaction of Skill Dependence with Ownership Concentration	−0.0414	−0.0721	−0.0100
	(0.381)	(0.132)	(0.210)
Equity Dependence	**0.0319***	0.0155	0.0180
	(0.052)	(0.118)	(0.112)
Financial Development	0.0453	0.0219	0.0344
	(0.303)	(0.220)	(0.328)
Skill Dependence	0.0018	−0.0063	−0.0022
	(0.403)	(0.389)	(0.419)
Ownership Concentration	0.0080	0.0053	0.0011
	(0.621)	(0.419)	(0.521)
Regression R^2−adj.	0.3601	0.3319	0.3680
Number of observations	7,389	7,389	7,389

This table reports the results of OLS regressions of the logs of total return volatility, idiosyncratic volatility, and systematic volatility on the interaction terms of industry contract enforcement sensitivity with national elections, autocracy, political risk, and control variables. All of the variables are defined in Table 1. Every regression includes industry, country, and year fixed effects. The numbers in parentheses are p-values. The coefficients significant at the 10% level (based on a two-tailed test) or higher are in boldface. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered by countries and years to adjust for heteroscedasticity, cross-sectional, and time-series correlations.

Table 6
Volatility of labor-intensive industries conditional on elections and party orientation

Specification	Total volatility 1	Idiosyncratic volatility 2	Systematic volatility 3
Interaction of Labor Intensity with Left Party Orientation	**0.1609***	0.0861	**0.1920***
	(0.004)	(0.416)	**(0.000)**
Interaction of Labor Intensity with National Elections	**0.0566***	0.0094	**0.0409***
	(0.080)	(0.358)	**(0.050)**
Labor Intensity	**0.1716***	0.1480	**0.1201***
	(0.000)	(0.211)	**(0.100)**
Left Party Orientation	−0.0220	**−0.0496***	0.0360
	(0.116)	**(0.028)**	(0.230)
National Elections	**0.1250***	0.1480	**0.1130***
	(0.081)	(0.140)	**(0.080)**
Interaction of Labor Intensity with GDP per Capita	−0.0262	−0.0240	**−0.1290***
	(0.202)	(0.321)	**(0.000)**
Interaction of Labor Intensity with Efficiency of Law	0.0228	0.0490	0.0172
	(0.434)	(0.182)	(0.504)
Interaction of Labor Intensity with Exchange Rate Risk	−0.0130	−0.0340	**−0.0866***
	(0.200)	(0.102)	**(0.030)**
GDP per Capita	**−0.2888***	**−0.2540***	**−0.2011***
	(0.000)	**(0.000)**	**(0.000)**
Efficiency of Law	−0.0086	−0.0203	−0.0042
	(0.662)	(0.320)	(0.714)
Exchange Rate Risk	**0.0320***	**0.0319***	0.0120
	(0.000)	**(0.000)**	(0.130)
Industry Size	**−0.0216***	**−0.0131***	−0.0036
	(0.000)	**(0.049)**	(0.303)
Industry leverage	**0.0707***	0.0080	**0.0600***
	(0.029)	(0.148)	**(0.000)**
Industry diversification	**−0.0279***	**−0.0414***	0.0041
	(0.000)	**(0.000)**	(0.280)
Interaction of Equity Dependence with Financial Development	−0.0830	−0.0777	−0.0803
	(0.231)	(0.308)	(0.250)
Interaction of Skill Dependence with Ownership Concentration	−0.0890	−0.0764	−0.0402
	(0.328)	(0.162)	(0.209)
Equity Dependence	**0.0403***	**0.0241***	**0.0202***
	(0.100)	**(0.082)**	**(0.100)**
Financial Development	0.0493	0.0222	0.0228
	(0.222)	(0.228)	(0.186)
Skill Dependence	0.0010	−0.0010	−0.0062
	(0.114)	(0.165)	(0.180)
Ownership Concentration	0.0062	0.0051	0.0012
	(0.513)	(0.444)	(0.612)
Regression R^2–adj.	0.3930	0.3814	0.3827
Number of observations	11,518	11,518	11,518

This table reports the results of OLS regressions of the logs of total return volatility, idiosyncratic volatility, and systematic volatility on the interaction terms of industry labor intensity with the left party orientation dummy variable, national elections dummy variable, and control variables. All of the variables are defined in Table 1. Every regression includes industry, country, and year fixed effects. The numbers in parentheses are p-values. The coefficients significant at the 10% level (based on a two-tailed test) or higher are in boldface. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered by countries and years to adjust for heteroscedasticity, cross-sectional, and time-series correlation.

5.7 Table 6: labor intensity, elections, and left-party government

Read:

- Labor-intensive industries have higher volatility under left-party governments.
- The interaction between labor intensity and left-party orientation is **0.1609** for total volatility and **0.1920** for systematic volatility, both highly significant.
- The interaction with national elections is positive and marginally significant for total and systematic volatility.
- The idiosyncratic-volatility coefficients are not statistically significant in the main specification.
- The table implies that the labor channel is mostly a systematic-risk channel, not a firm-specific risk channel.
- The interpretation is that labor-market policy and labor adjustment costs affect broad cash-flow exposure for labor-intensive sectors.

5.8 Table 7: decomposition of left-party orientation

Read:

- Table 7 addresses the concern that left-party governments are elected after poor economic outcomes.
- The authors first predict left-party orientation using lagged macro variables, then split party orientation into explained and unexplained components.
- The interaction of labor intensity with the unexplained component remains positive and significant for all three volatility measures.
- This supports the view that the labor-party result is not simply driven by past volatility or macroeconomic performance.

5.9 Table 8: labor intensity and employment rigidity

Read:

- Table 8 uses a more direct labor-regulation measure: the World Bank rigidity of employment index.
- The interaction between labor intensity and employment rigidity is positive for total volatility and systematic volatility.
- The coefficients are **0.0581** for total volatility and **0.0778** for systematic volatility.
- This table strengthens the labor-channel interpretation because it uses a direct measure of employment regulation rather than only party orientation.

5.10 Robustness tests

Read:

- The results survive instrumental-variable specifications for autocracy and political risk.

Table 7
Decomposition of left party orientation variable

Specification	Total volatility 1	Idiosyncratic volatility 2	Systematic volatility 3
Interaction of Labor Intensity with Unexplained Portion of Left Party Orientation	**0.0682*****	**0.0590*****	**0.0190****
	(0.000)	**(0.000)**	**(0.000)**
Interaction of Labor Intensity with Explained Portion of Left Party Orientation	**0.0530***	**0.0232****	**0.0617***
	(0.052)	**(0.030)**	**(0.100)**
Labor Intensity	**0.3082*****	0.2886	0.2089
	(0.000)	(0.144)	(0.117)
Unexplained Portion of Left Party Orientation	−0.0218	−0.0421	−0.0406
	(0.176)	(0.187)	(0.414)
Explained Portion of Left Party Orientation	−0.0030	−0.005	−0.0030
	(0.608)	(0.402)	(0.180)
Interaction of Labor Intensity with GDP per Capita	−0.0186	−0.0128	−0.0211
	(0.208)	(0.302)	(0.128)
Interaction of Labor Intensity with Efficiency of Law	0.0267	0.0308	0.0161
	(0.311)	(0.249)	(0.202)
Interaction of Labor Intensity with Exchange Rate Risk	**−0.0246*****	**−0.0208*****	**−0.0462*****
	(0.000)	**(0.000)**	**(0.000)**
GDP per Capita	**−0.3060*****	**−0.3240*****	**−0.4221*****
	(0.000)	**(0.000)**	**(0.000)**
Efficiency of Law	−0.0882	−0.0940	−0.0050
	(0.209)	(0.338)	(0.725)
Exchange Rate Risk	**0.0305*****	**0.0338*****	0.0106
	(0.000)	**(0.000)**	(0.300)
Industry Size	**−0.0265*****	**−0.0110*****	−0.0082
	(0.000)	**(0.000)**	(0.160)
Industry Leverage	**0.0932*****	0.0012	**0.0800*****
	(0.000)	(0.364)	**(0.000)**
Industry Diversification	**−0.0336*****	**−0.0431*****	0.00304
	(0.000)	**(0.000)**	(0.256)
Interaction of Equity Dependence with Financial Development	−0.0345	−0.0360	−0.0351
	(0.350)	(0.241)	(0.238)
Interaction of Skill Dependence with Ownership Concentration	−0.0630	−0.0650	−0.0711
	(0.236)	(0.311)	(0.308)
Equity Dependence	0.0381	0.0271	0.0170
	(0.108)	(0.111)	(0.210)
Financial Development	0.0424	0.0293	0.0350
	(0.211)	(0.189)	(0.220)
Skill Dependence	0.0013	−0.0062	0.0041
	(0.482)	(0.311)	(0.339)
Ownership Concentration	0.0182	0.0191	−0.0030
	(0.409)	(0.330)	(0.208)
Regression R^2-adj.	0.3380	0.3318	0.3640
Number of observations	11,518	11,518	11,518

This table reports the results of OLS regressions of the logs of return volatility, idiosyncratic volatility, and systematic volatility on the interaction terms of labor intensity with unexplained and explained (by lagged macroeconomic variables) portions of party orientation and control variables. All of the variables are defined in Table 1. Every regression includes industry, country, and year fixed effects. The numbers in parentheses are p-values. The coefficients significant at the 10% level (based on a two-tailed test) or higher are in boldface. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered by countries and years to adjust for heteroscedasticity, cross-sectional, and time-series correlations.

Table 8
Volatility of labor-intensive industries conditional on rigidity of employment index

Specification	Total volatility 1	Idiosyncratic volatility 2	Systematic volatility 3
Interaction of Labor Intensity with Rigidity of Employment Index	**0.0581***	0.0230	**0.0778****
	(0.052)	(0.316)	**(0.031)**
Rigidity of Employment Index	**0.5630*****	**0.7181*****	**0.7310*****
	(0.000)	**(0.000)**	**(0.000)**
Labor Intensity	**0.1902***	0.0702	0.1212
	(0.100)	(0.288)	(0.306)
Interaction of Labor Intensity with GDP per Capita	0.1211	0.1621	−0.1714
	(0.262)	(0.120)	(0.493)
Interaction of Labor Intensity with Efficiency of Law	0.0386	0.0312	0.0514
	(0.314)	(0.110)	(0.229)
Interaction of Labor Intensity with Exchange Rate Risk	**−0.0208****	**−0.0301***	**−0.0560****
	(0.020)	**(0.100)**	**(0.041)**
GDP per Capita	**−0.2359*****	**−0.2903*****	**−0.2000*****
	(0.000)	**(0.000)**	**(0.000)**
Efficiency of Law	−0.0160	−0.0414	−0.0190
	(0.336)	(0.421)	(0.403)
Exchange Rate Risk	**0.0342*****	**0.0338*****	**0.0316*****
	(0.000)	**(0.000)**	**(0.000)**
Industry Size	**−0.0104*****	**−0.0081****	−0.0039
	(0.000)	**(0.030)**	(0.308)
Industry Leverage	**0.0516*****	0.0014	**0.0816*****
	(0.000)	(0.420)	**(0.000)**
Industry Diversification	**−0.0220*****	**−0.0216*****	0.0099
	(0.000)	**(0.000)**	(0.214)
Interaction of Equity Dependence with Financial Development	**−0.1488***	−0.0562	**−0.1405***
	(0.100)	(0.216)	**(0.100)**
Interaction of Skill Dependence with Ownership Concentration	0.0016	0.0010	0.0119
	(0.312)	(0.409)	(0.365)
Equity Dependence	0.0191	0.0152	0.0173
	(0.190)	(0.121)	(0.314)
Financial Development	0.0590	0.0120	0.0454
	(0.286)	(0.134)	(0.180)
Skill Dependence	−0.0031	−0.0030	0.0042
	(0.518)	(0.431)	(0.615)
Ownership Concentration	0.0231	0.0011	0.0060
	(0.122)	(0.214)	(0.328)
Regression R^2–adj.	0.2693	0.2560	0.2822
Number of observations	2,005	2,005	2,005

This table reports the results of OLS regressions of the logs of total return volatility, idiosyncratic volatility, and systematic volatility on the interaction term of industry labor intensity with country rigidity of employment index and control variables. The regressions are run using the panel of industry-country-year observations from 2004 to 2006. Every regression includes country, industry, and year fixed effects. All of the variables are defined in Table 1. The numbers in parentheses are p-values. The coefficients significant at the 10% level (based on a two-tailed test) or higher are in boldface. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered by countries and years to adjust for heteroscedasticity, cross-sectional, and time-series correlations.

- They are not driven by early elections or flexible election systems.
- They survive controls for political crises and financial crises.
- They survive alternative volatility construction using size, debt, or diversification weights.
- They survive restricting the sample to pure-play firms.
- They survive controlling for ROA volatility, suggesting that market volatility is not merely a mechanical reflection of fundamental volatility.
- They survive using country-specific labor intensity from Worldscope.
- They survive including the three political sensitivities jointly, though the labor-intensity channel becomes somewhat less significant when all channels are included together.

6 Short summary

- **Method.** The paper uses a cross-country, cross-industry panel design that interacts industry political sensitivities with country political variables.
- **Measurement.** Political sensitivities are trade exposure, contract enforcement sensitivity, and labor intensity. Political variables include elections, autocracy, political risk, left-party orientation, and employment rigidity.

- **Main result.** Industries with higher political exposure become more volatile when the relevant political risk is high.
- **Domestic versus global risk.** Domestic political risk mainly affects systematic volatility, while foreign political risk transmitted through trade mainly affects idiosyncratic volatility.
- **Economic implication.** International trade can transmit foreign political risk, and industry input structure and labor dependence shape the volatility cost of political uncertainty.

7 Conclusion

7.1 Core conclusion

- The paper shows that political risk affects stock-return volatility, but not uniformly across industries.
- The transmission channels are economically intuitive: trade exposure, contract enforcement dependence, and labor intensity.
- Routine political events, such as elections and shifts in party orientation, can matter for financial volatility even outside dramatic crises.

7.2 Economic intuition

- Export-oriented industries are exposed to domestic trade-policy uncertainty and to political conditions in trading-partner countries.
- Contract-dependent industries are exposed to political shocks that affect institutional quality and the reliability of supplier relationships.
- Labor-intensive industries are exposed to political shifts in labor regulation and employment protection.
- Domestic political risk affects broad market factors and therefore systematic volatility, while foreign political shocks through trade are harder to diversify fully and appear more in idiosyncratic volatility.

7.3 Takeaway

This paper turns the broad idea that "politics affects markets" into a structured industry-level empirical design. The key lesson is that political risk is transmitted through specific economic exposures. A firm's industry determines whether political events show up mainly as systematic risk, idiosyncratic risk, or both. For investors, the paper implies that political-risk management should account for industry trade links, input complexity, and labor dependence. For firms, it implies that decisions about exporting, supply-chain complexity, and labor-intensive production can affect the volatility of cash flows and the cost of capital.